PhenoCam vs model transition dates, PhenoCam BDT pixels, 30-d nowcast windows anchored on the observed LCSIF half-amplitude date

LCSIF window (observed)

n = 508 site-yr, bias +3 d, MAE 9 d

within-site r = 0.55
49 sites ≥5 y, slope 0.47

n = 547 site-yr, bias -7 d, MAE 15 d

within-site r = 0.04
56 sites ≥5 y, slope 0.01

TFT

n = 507 site-yr, bias +3 d, MAE 9 d

within-site r = 0.54
49 sites ≥5 y, slope 0.40

n = 547 site-yr, bias -7 d, MAE 15 d

within-site r = 0.02
56 sites ≥5 y, slope 0.01

PhenoFormerSeq

n = 508 site-yr, bias +2 d, MAE 9 d

within-site r = 0.56
49 sites ≥5 y, slope 0.43

n = 547 site-yr, bias -6 d, MAE 15 d

within-site r = -0.01
56 sites ≥5 y, slope -0.00

4GST (OLS extrap.)

n = 66 site-yr, bias +9 d, MAE 14 d
87% of forecasts never cross

no site with ≥5 matched years
(too many forecasts never cross)

n = 376 site-yr, bias +0 d, MAE 15 d
31% of forecasts never cross

within-site r = 0.03
34 sites ≥5 y, slope 0.02

SOS: model date (DOY)
PhenoCam date (DOY)
SOS: model anomaly (d)
PhenoCam anomaly (d)
EOS: model date (DOY)
PhenoCam date (DOY)
EOS: model anomaly (d)
PhenoCam anomaly (d)

• test period 2008–10 ▲ train period ≥2011 (in-sample)

2011–2021 lie in the training years of the TFT and PhenoFormerSeq checkpoints (in-sample); only 2008–2010 are test years. LSTM omitted (categorical embeddings not saved).